3.2 Functional analysis: solutions

Read a solution only after a serious attempt. After reading, close the file and write the proof out yourself.

Warm-up

1.

Completeness of ℓ^∞. Let $(x^{(k)})$ be Cauchy in ℓ^∞. For each coordinate n, $|x_n^{(k)} - x_n^{(j)}| \le \|x^{(k)} - x^{(j)}\|_\infty$, so $(x_n^{(k)})_k$ is Cauchy in $\mathbb{C}$; let x_n be its limit. Given $\varepsilon > 0$ choose N with $\|x^{(k)} - x^{(j)}\|_\infty < \varepsilon$ for $j, k \ge N$. For each n and $k \ge N$, letting $j \to \infty$ gives $|x_n^{(k)} - x_n| \le \varepsilon$. So $x = x^{(N)} + (x - x^{(N)})$ is bounded, i.e. $x \in \ell^\infty$, and $\|x^{(k)} - x\|_\infty \le \varepsilon$ for $k \ge N$.

c_0 is closed. Let $x^{(k)} \in c_0$ with $x^{(k)} \to x$ in ℓ^∞. Given ε, pick k with $\|x^{(k)} - x\|_\infty < \varepsilon/2$, then N with $|x_n^{(k)}| < \varepsilon/2$ for $n \ge N$. Then $|x_n| < \varepsilon$ for $n \ge N$, so $x \in c_0$. (It is clearly a subspace.)

2.

(a) Linearity is clear. Let $f_n(x) = (1-x)^n$. Then $\|f_n\|_\infty = 1$ and $f_n'(0) = -n$, so $|\varphi(f_n)|/\|f_n\|_\infty = n \to \infty$.

(b) Let $\dim X = d$ with basis $v_1, \dots, v_d$, and define $\|\sum c_i v_i\|_1 = \sum |c_i|$. This is a norm on X, so by equivalence there is A with $\|x\|_1 \le A\|x\|$ for all x. For linear $T : X \to Y$ let $B = \max_i \|Tv_i\|$. Then

$$\Big\|T \sum c_i v_i\Big\| \le \sum |c_i| \|Tv_i\| \le B\|x\|_1 \le AB\|x\|.$$

3.

$\|Mf\|_2^2 = \int_0^1 x^2 |f|^2 \le \int_0^1 |f|^2$, so $\|M\| \le 1$. For $f_\varepsilon = \varepsilon^{-1/2} \chi_{[1-\varepsilon, 1]}$ we have $\|f_\varepsilon\|_2 = 1$ and $\|Mf_\varepsilon\|_2^2 = \varepsilon^{-1} \int_{1-\varepsilon}^1 x^2\,dx \ge (1-\varepsilon)^2$. So $\|M\| = 1$.

If $\|Mf\|_2 = \|f\|_2$ then $\int_0^1 (1 - x^2)|f(x)|^2\,dx = 0$. The integrand is non-negative, so it vanishes a.e.; since $1 - x^2 > 0$ on $[0, 1)$, $f = 0$ a.e.

4.

If a norm comes from an inner product it satisfies the parallelogram law. Take $x = e_1$, $y = e_2$. For $p < \infty$, $\|x + y\|_p = \|x - y\|_p = 2^{1/p}$, so the law reads $2 \cdot 2^{2/p} = 4$, i.e. $2^{2/p} = 2$, i.e. $p = 2$. For $p = \infty$ the left side is $1 + 1 = 2 \ne 4$. Conversely, ℓ^2 has the inner product $\sum x_n \bar{y}_n$.

5.

The closest point $g = a + bx$ is the orthogonal projection, characterised by $x^2 - g \perp 1$ and $x^2 - g \perp x$:

$$\int_0^1 (x^2 - a - bx)\,dx = \tfrac{1}{3} - a - \tfrac{b}{2} = 0, \qquad \int_0^1 x(x^2 - a - bx)\,dx = \tfrac{1}{4} - \tfrac{a}{2} - \tfrac{b}{3} = 0.$$

From the first, $a = \frac{1}{3} - \frac{b}{2}$. Substituting, $\frac{1}{4} - \frac{1}{6} + \frac{b}{4} - \frac{b}{3} = 0$, i.e. $\frac{1}{12} - \frac{b}{12} = 0$, so $b = 1$ and $a = -\frac{1}{6}$. The closest point is $g(x) = x - \frac{1}{6}$.

The residual $r = x^2 - x + \frac{1}{6}$ is orthogonal to g, so $\|r\|^2 = \langle r, x^2\rangle = \int_0^1 \left(x^4 - x^3 + \frac{x^2}{6}\right) dx = \frac{1}{5} - \frac{1}{4} + \frac{1}{18} = \frac{1}{180}$. The distance is $\frac{1}{\sqrt{180}} = \frac{1}{6\sqrt{5}}$.

6.

(a) Let $x_n \rightharpoonup x$. Consider $J(x_n) \in X^{**} = B(X^*, \mathbb{C})$. For each $f \in X^*$, $(f(x_n))$ converges, so $\sup_n |J(x_n)(f)| < \infty$. X^* is a Banach space, so the uniform boundedness principle gives $\sup_n \|J(x_n)\| < \infty$. Since J is an isometry (Hahn–Banach), $\sup_n \|x_n\| < \infty$.

(b) By the Riesz representation theorem every $f \in H^*$ has the form $f = \langle \cdot, y\rangle$. Bessel's inequality gives $\sum_n |\langle e_n, y\rangle|^2 \le \|y\|^2 < \infty$, so $\langle e_n, y\rangle \to 0$. Hence $e_n \rightharpoonup 0$. But $\|e_n - 0\| = 1$ for all n.

Core

7.

(a) Take $x_0 \notin M$ and let $d = \operatorname{dist}(x_0, M)$; $d > 0$ because M is closed. Since $d/\theta > d$, there is $m_0 \in M$ with $\|x_0 - m_0\| < d/\theta$. Put $x = \frac{x_0 - m_0}{\|x_0 - m_0\|}$. For any $m \in M$, $m_0 + \|x_0 - m_0\| m \in M$, so

$$\|x - m\| = \frac{\|x_0 - (m_0 + \|x_0 - m_0\| m)\|}{\|x_0 - m_0\|} \ge \frac{d}{\|x_0 - m_0\|} > \theta.$$

(b) If $\dim X = d < \infty$, choosing a basis gives a linear bijection $\mathbb{F}^d \to X$ which is a homeomorphism (equivalence of norms); the closed unit ball is closed and bounded in $\mathbb{F}^d$-coordinates, hence compact by Heine–Borel.

If X is infinite-dimensional, build unit vectors inductively: given $x_1, \dots, x_{n-1}$, the span M_{n-1} is finite-dimensional, hence closed, and proper. By (a) with $\theta = \frac{1}{2}$ choose a unit x_n with $\operatorname{dist}(x_n, M_{n-1}) \ge \frac{1}{2}$. Then $\|x_n - x_m\| \ge \frac{1}{2}$ for $m < n$, so (x_n) has no Cauchy subsequence, and the unit ball is not (sequentially) compact.

8.

(a) If $x = 0$ take $f = 0$. Otherwise, on $M = \mathbb{F}x$ define $g(\lambda x) = \lambda \|x\|$; $\|g\| = 1$. By Hahn–Banach extend to $f \in X^*$ with $\|f\| = 1$.

(b) Let $\{f_n\}$ be dense in X^*. Choose unit vectors x_n with $|f_n(x_n)| \ge \frac{1}{2}\|f_n\|$. Let M be the closed linear span of $\{x_n\}$; it is separable (finite rational linear combinations are dense). Suppose $M \ne X$. By Hahn–Banach (corollary 3 in the notes) there is $f \in X^*$ with $\|f\| = 1$ and $f|_M = 0$. Pick n with $\|f - f_n\| < \frac{1}{3}$. Then $\|f_n\| > \frac{2}{3}$, and

$$\tfrac{1}{3} < \tfrac{1}{2}\|f_n\| \le |f_n(x_n)| = |(f_n - f)(x_n)| \le \|f_n - f\| < \tfrac{1}{3},$$

a contradiction. So $X = M$ is separable.

9.

For $b \in \ell^\infty$ and $a \in \ell^1$, $|\sum a_n b_n| \le \|b\|_\infty \|a\|_1$, so $\varphi_b \in (\ell^1)^*$ with $\|\varphi_b\| \le \|b\|_\infty$. Also $\varphi_b(e_n) = b_n$ and $\|e_n\|_1 = 1$, so $\|\varphi_b\| \ge \sup_n |b_n|$. Hence $b \mapsto \varphi_b$ is a linear isometry.

Onto. Let $\varphi \in (\ell^1)^*$ and $b_n = \varphi(e_n)$. Then $|b_n| \le \|\varphi\|$, so $b \in \ell^\infty$. For $a \in \ell^1$, $\|a - \sum_{n \le N} a_n e_n\|_1 = \sum_{n > N} |a_n| \to 0$, so by continuity and linearity $\varphi(a) = \lim_N \sum_{n \le N} a_n b_n = \varphi_b(a)$.

10.

We use the closed graph theorem (H is complete). Suppose $x_n \to x$ and $Tx_n \to z$. For every $y \in H$,

$$\langle z, y\rangle = \lim\langle Tx_n, y\rangle = \lim\langle x_n, Ty\rangle = \langle x, Ty\rangle = \langle Tx, y\rangle.$$

So $z - Tx \perp H$, i.e. $z = Tx$. The graph is closed, so T is bounded.

11.

(a) The identity map $(V, \|\cdot\|_2) \to (V, \|\cdot\|_1)$ is linear, bijective and bounded (by C) between Banach spaces. By the bounded inverse theorem, its inverse is bounded: there is C' with $\|v\|_2 \le C'\|v\|_1$.

(b) $f_n(x) = x^n$ has $\|f_n\|_\infty = 1$ and $\|f_n\|_{L^1} = \frac{1}{n+1} \to 0$, so there is no C' with $\|f\|_\infty \le C'\|f\|_{L^1}$. $(C[0,1], \|\cdot\|_\infty)$ is complete. If $(C[0,1], \|\cdot\|_{L^1})$ were complete, (a) would make the norms equivalent. So it is not complete.

12.

For $N \ge 1$ define $\varphi_N \in c_0^*$ by $\varphi_N(b) = \sum_{n\le N} a_n b_n$. Then $|\varphi_N(b)| \le \left(\sum_{n\le N}|a_n|\right)\|b\|_\infty$, and choosing $b_n = \overline{\operatorname{sgn} a_n}$ for $n \le N$ and $b_n = 0$ for $n > N$ (a finitely supported sequence, so in c_0, with $\|b\|_\infty \le 1$) shows $\|\varphi_N\| = \sum_{n\le N}|a_n|$. (Here $\operatorname{sgn} z = z/|z|$ for $z \ne 0$ and 0 otherwise.)

For each fixed $b \in c_0$, $\varphi_N(b)$ converges as $N \to \infty$ by hypothesis, so $\sup_N|\varphi_N(b)| < \infty$. Since c_0 is a Banach space (Problem 1), the uniform boundedness principle gives $\sup_N\|\varphi_N\| < \infty$, that is, $\sum_n|a_n| < \infty$.

13.

Let $\|x_n\| \le C$. Let M be the closed linear span of $\{x_n\}$.

Diagonal argument. For each j, the sequence $(\langle x_j, x_n\rangle)_n$ is bounded by C^2. By a diagonal argument choose a subsequence $(y_k) = (x_{n_k})$ such that $\lim_k\langle z, y_k\rangle$ exists for every $z \in \{x_1, x_2, \dots\}$, hence for every z in their linear span S.

Extend to H. If $z \in M = \overline{S}$ and $\varepsilon > 0$, choose $s \in S$ with $\|z - s\| < \varepsilon$. Then $|\langle z, y_k\rangle - \langle z, y_l\rangle| \le 2C\varepsilon + |\langle s, y_k\rangle - \langle s, y_l\rangle|$, so $(\langle z, y_k\rangle)_k$ is Cauchy. If $z \in M^\perp$, then $\langle z, y_k\rangle = 0$. Since $H = M \oplus M^\perp$, $\varphi(z) = \lim_k\langle z, y_k\rangle$ exists for every $z \in H$.

Identify the limit. φ is linear and $|\varphi(z)| \le C\|z\|$. By Riesz representation, $\varphi(z) = \langle z, y\rangle$ for some $y \in H$. Thus $\langle z, y_k\rangle \to \langle z, y\rangle$ for all z; taking complex conjugates, $\langle y_k, z\rangle \to \langle y, z\rangle$. Since every functional is of the form $\langle \cdot, z\rangle$, $y_k \rightharpoonup y$.

14.

Weak closure of S is contained in B. $B = \bigcap_{\|z\|\le 1}\{x : \operatorname{Re}\langle x, z\rangle \le 1\}$ (because $\|x\| = \sup_{\|z\|\le 1}\operatorname{Re}\langle x, z\rangle$). Each set in the intersection is weakly closed (preimage of a closed set under a weakly continuous map), so B is weakly closed and contains S.

B is contained in the weak closure of S. Points of S are trivially in it. Let $\|x\| < 1$. The subspace $\{x\}^\perp$ is infinite-dimensional (it has codimension at most 1 in an infinite-dimensional space), so it contains an orthonormal sequence (e_n). Put $t = \sqrt{1 - \|x\|^2} > 0$ and $y_n = x + te_n$. By Pythagoras $\|y_n\|^2 = \|x\|^2 + t^2 = 1$, so $y_n \in S$. By Problem 6(b), $y_n \rightharpoonup x$. A weak limit of a sequence in S lies in the weak closure of S (every weak neighbourhood of x contains y_n for large n). So $x \in \overline{S}^w$.

15.

($\Leftarrow$) Let D_N be the finite-rank operator $D_N e_n = \lambda_n e_n$ for $n \le N$ and 0 for $n > N$. Then $(D - D_N)$ is diagonal with entries $0, \dots, 0, \lambda_{N+1}, \lambda_{N+2}, \dots$, so $\|D - D_N\| = \sup_{n>N} |\lambda_n| \to 0$. A norm limit of finite-rank operators is compact.

($\Rightarrow$) If $\lambda_n \not\to 0$, there are $\varepsilon > 0$ and $n_1 < n_2 < \cdots$ with $|\lambda_{n_k}| \ge \varepsilon$. The sequence e_{n_k} is bounded, but for $k \ne l$, $\|De_{n_k} - De_{n_l}\|^2 = |\lambda_{n_k}|^2 + |\lambda_{n_l}|^2 \ge 2\varepsilon^2$, so (De_{n_k}) has no convergent subsequence. D is not compact.

16.

Compact. V is the integral operator with kernel $k(x,t) = \chi_{\{t<x\}}$, which is in $L^2([0,1]^2)$. Hilbert–Schmidt integral operators are compact.

No eigenvalues. Suppose $Vf = \lambda f$ with $f \in L^2$.

- If $\lambda = 0$: $\int_0^x f = 0$ for all x. By the Lebesgue differentiation theorem, $f(x) = \frac{d}{dx}\int_0^x f = 0$ for a.e. x. So $f = 0$.
- If $\lambda \ne 0$: $f = \lambda^{-1} Vf$ is continuous (as Vf is), hence Vf is C^1, hence f is C^1, with $f' = \lambda^{-1} f$ and $f(0) = \lambda^{-1}(Vf)(0) = 0$. The unique solution of this initial value problem is $f = 0$.

Spectrum. $L^2[0,1]$ is infinite-dimensional, so the compact operator V is not invertible, and $0 \in \sigma(V)$. For a compact operator every non-zero point of the spectrum is an eigenvalue (Riesz–Schauder). There are none, so $\sigma(V) = \{0\}$.

(Alternative: by induction $|(V^n f)(x)| \le \frac{x^{n-1}}{(n-1)!}\|f\|_1$, which gives $\|V^n\| \le \frac{1}{(n-1)!}$ and so spectral radius $\lim \|V^n\|^{1/n} = 0$.)

17.

The kernel is real, symmetric and in L^2, so K is compact and self-adjoint. Explicitly

$$(Kf)(x) = \int_0^x y f(y)\, dy + x \int_x^1 f(y)\, dy.$$

For $f \in L^2 \subseteq L^1$, $u = Kf$ is absolutely continuous with

$$u'(x) = x f(x) + \int_x^1 f - x f(x) = \int_x^1 f(y)\, dy \quad \text{a.e.},$$

and u' (which has a continuous version) is itself absolutely continuous with $u'' = -f$ a.e. Also $u(0) = 0$ and $u'(1) = 0$.

Kernel. If $Kf = 0$ then $u'(x) = \int_x^1 f = 0$ for all x, so $f = 0$ a.e. (Lebesgue differentiation). So $\ker K = 0$.

Positivity. $\min(x,y) = \int_0^1 \chi_{[0,x)}(t)\chi_{[0,y)}(t)\, dt$, so by Fubini $\langle Kf, f\rangle = \int_0^1 \left|\int_t^1 f(x)\, dx\right|^2 dt \ge 0$. Hence all eigenvalues are ≥ 0, and since $\ker K = 0$ they are > 0.

Eigenvalues. Let $Kf = \lambda f$ with $\lambda > 0$, $f \ne 0$. Then $f = \lambda^{-1} u$ is continuous, hence (bootstrapping) smooth, and $\lambda f'' = -f$, $f(0) = 0$, $f'(1) = 0$. With $\omega = \lambda^{-1/2}$, $f = A \sin \omega x$ (the condition $f(0) = 0$ kills the cosine), and $f'(1) = A\omega \cos \omega = 0$ forces $\cos \omega = 0$, i.e. $\omega = (n - \frac{1}{2})\pi$, $n \ge 1$.

Conversely each such f satisfies $Kf = \lambda f$: $u = Kf$ and λf both solve $v'' = -f$, $v(0) = 0$, $v'(1) = 0$, and the difference w satisfies $w'' = 0$, $w(0) = 0$, $w'(1) = 0$, so $w = 0$. So

$$\lambda_n = \frac{1}{(n-\frac{1}{2})^2\pi^2}, \qquad f_n(x) = \sqrt{2}\sin\left((n-\tfrac{1}{2})\pi x\right), \quad n = 1, 2, ...$$

Each eigenspace is one-dimensional (the ODE solution is unique up to scaling). Normalisation: $\int_0^1 \sin^2(\omega x)\,dx = \frac{1}{2} - \frac{\sin 2\omega}{4\omega} = \frac{1}{2}$ since $\sin 2\omega = \sin((2n-1)\pi) = 0$.

Basis. By the spectral theorem, $L^2[0,1] = \ker K \oplus \overline{\text{span}}\{f_n\} = \overline{\text{span}}\{f_n\}$. Eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal. So $\{f_n\}$ is an orthonormal basis.

(Check: the trace $\sum \lambda_n = \frac{4}{\pi^2}\sum_{n\geq 1}\frac{1}{(2n-1)^2} = \frac{4}{\pi^2}\cdot\frac{\pi^2}{8} = \frac{1}{2}$ equals $\int_0^1 k(x,x)\,dx = \int_0^1 x\,dx = \frac{1}{2}$.)

18.

$y \in \ker T^* \iff T^*y = 0 \iff \langle x, T^*y\rangle = 0\ \forall x \iff \langle Tx, y\rangle = 0\ \forall x \iff y \in (\operatorname{ran} T)^\perp$.

For the second identity, take orthogonal complements: $(\ker T^*)^\perp = (\operatorname{ran} T)^{\perp\perp} = \overline{\operatorname{ran} T}$, using $S^{\perp\perp} = \overline{S}$ for a subspace S (which follows from $H = \overline{S} \oplus \overline{S}^\perp$ and $S^\perp = \overline{S}^\perp$).

Qualifying-exam level

19.

Let $N = \ker T$.

Step 0: N is finite-dimensional. On N, $K = I$. The unit ball of N equals $K(\text{unit ball of } N)$, which is relatively compact; it is also closed. So the unit ball of N is compact and $\dim N < \infty$ (Problem 7).

(a) Let $Tu_n \to f$. Replace u_n by $u_n - P_N u_n$, so $u_n \in N^\perp$ (this does not change Tu_n).

Claim: (u_n) is bounded. If not, pass to a subsequence with $\|u_n\| \to \infty$ and set $v_n = u_n/\|u_n\| \in N^\perp$. Then $Tv_n \to 0$. By compactness a further subsequence has $Kv_n \to w$. Then $v_n = Tv_n + Kv_n \to w$, so $\|w\| = 1$, $w \in N^\perp$ (closed), and $Tw = \lim Tv_n = 0$, i.e. $w \in N$. So $w \in N \cap N^\perp = \{0\}$, contradicting $\|w\| = 1$.

Now (u_n) is bounded, so a subsequence has $Ku_n \to z$. Then $u_n = Tu_n + Ku_n \to f + z =: u$, and $Tu = \lim Tu_n = f$. So $f \in \operatorname{ran} T$, and the range is closed.

(b) Suppose T is injective but not surjective. Let $H_n = T^n H$. Note $T^n = (I-K)^n = I - K_n$ where K_n is compact (expand by the binomial theorem; every term except I contains K). By (a) applied to K_n, each H_n is closed. Clearly $H_0 \supseteq H_1 \supseteq H_2 \supseteq \cdots$. The inclusions are strict: $H_1 \neq H_0$ by assumption, and if $H_{n+1} = H_n$ for some $n \geq 1$, then for any x we have $T^n x = T^{n+1} y$ for some y, so by injectivity of T^n, $x = Ty$, i.e. T is onto, a contradiction.

Choose unit vectors $x_n \in H_n \cap H_{n+1}^\perp$. For $n < m$,

$$Kx_n - Kx_m = x_n - (Tx_n + x_m - Tx_m)\,.$$

Here $Tx_n \in H_{n+1}$, $x_m \in H_m \subseteq H_{n+1}$ and $Tx_m \in H_{m+1} \subseteq H_{n+1}$, so the bracket z lies in H_{n+1}, while $x_n \perp H_{n+1}$. By Pythagoras $\|Kx_n - Kx_m\|^2 = \|x_n\|^2 + \|z\|^2 \geq 1$. So (Kx_n) has no convergent subsequence although (x_n) is bounded, contradicting compactness of K. Hence T is surjective.

20.

Bounded below. By Cauchy–Schwarz, $c\|x\|^2 \le \operatorname{Re}\langle Tx, x\rangle \le \|Tx\|\|x\|$, so $\|Tx\| \ge c\|x\|$. In particular T is injective.

Closed range. If $Tx_n \to y$, then $\|x_n - x_m\| \le c^{-1}\|Tx_n - Tx_m\|$, so (x_n) is Cauchy, $x_n \to x$, and $Tx = y$.

Dense range. If $y \perp \operatorname{ran} T$ then $0 = \operatorname{Re}\langle Ty, y\rangle \ge c\|y\|^2$, so $y = 0$. Thus $(\operatorname{ran} T)^\perp = 0$ and the range is dense.

So T is a bijection. For $y = Tx$, $\|T^{-1}y\| = \|x\| \le c^{-1}\|Tx\| = c^{-1}\|y\|$, so $\|T^{-1}\| \le 1/c$.

(This is the operator form of the Lax–Milgram theorem: apply it to the operator representing a bounded coercive sesquilinear form $a(u, v) = \langle Tu, v\rangle$.)

21.

(a) Let $\{d_1, d_2, \dots\}$ be dense in H. Apply Gram–Schmidt to this sequence, discarding any vector that is a linear combination of earlier ones. The result is an orthonormal sequence (e_n) whose linear span equals the span of the d_j, hence is dense. Since H is infinite-dimensional, the sequence is infinite. A dense span means (e_n) is an orthonormal basis, so Parseval holds: $\|x\|^2 = \sum|\langle x, e_n\rangle|^2$.

Define $U : H \to \ell^2$ by $Ux = (\langle x, e_n\rangle)_n$. It is linear and, by Parseval, isometric. It is onto: given $a \in \ell^2$, the partial sums $s_N = \sum_{n\le N} a_n e_n$ satisfy $\|s_N - s_M\|^2 = \sum_{M<n\le N}|a_n|^2$, so they are Cauchy; the limit x has $\langle x, e_n\rangle = a_n$ by continuity of the inner product. An isometric linear bijection preserves inner products (polarisation).

(b) If $\{e_\alpha\}_{\alpha\in A}$ is orthonormal with A uncountable, then $\|e_\alpha - e_\beta\| = \sqrt{2}$ for $\alpha \ne \beta$. The open balls $B(e_\alpha, \frac{\sqrt{2}}{2})$ are pairwise disjoint. A dense subset must meet each of them, so it is uncountable.

22.

Recall the fact from Problem 8(b): if Y^* is separable then Y is separable. Apply it with $Y = X^*$: if X^{**} is separable, then X^* is separable.

(a) Under $c_0^* \cong \ell^1$ and $(\ell^1)^* \cong \ell^\infty$, the canonical map $J : c_0 \to c_0^{**}$ becomes the inclusion $c_0 \hookrightarrow \ell^\infty$: for $x \in c_0$ and $b \in \ell^1$, $J(x)(b) = \sum_n b_n x_n = \varphi_x(b)$. The sequence $(1, 1, 1, \dots) \in \ell^\infty$ is not in c_0, so J is not onto and c_0 is not reflexive.

(b) ℓ^1 is separable (finitely supported rational sequences are dense). $(\ell^1)^* \cong \ell^\infty$ is not separable: the uncountably many 0–1 sequences are at mutual distance 1, so (as in Problem 21(b)) no countable set is dense. If ℓ^1 were reflexive, $(\ell^1)^{**} = J(\ell^1)$ would be separable, and then $(\ell^1)^*$ would be separable by the fact above. Contradiction.

(c) Same argument. $L^1[0, 1]$ is separable (step functions with rational data are dense). $L^\infty[0, 1]$ is not separable: the functions $\chi_{[0,t]}$, $t \in (0, 1]$, satisfy $\|\chi_{[0,t]} - \chi_{[0,s]}\|_\infty = 1$ for $s \ne t$. If $L^1[0, 1]$ were reflexive, its bidual would be separable, hence its dual L^∞ would be separable. Contradiction.

23.

Write $R = \operatorname{ran} P$ and $N = \ker P$. Since $P^2 = P$, every x decomposes as $x = Px + (x - Px)$ with $Px \in R$ and $x - Px \in N$, and $Pu = u$ for $u \in R$. Also $\|P\| \ge 1$: pick $u \in R$, $u \ne 0$ (possible as $P \ne 0$); then $Pu = u$.

(i) ⇒ (ii). For $x, y \in H$: $\langle Px, y\rangle = \langle Px, Py + (y - Py)\rangle = \langle Px, Py\rangle$ because $Px \in R \perp N \ni y - Py$. Similarly $\langle x, Py\rangle = \langle Px, Py\rangle$. So $P = P^*$.

(ii) ⇒ (i). For $u = Px \in R$ and $v \in N$: $\langle u, v\rangle = \langle Px, v\rangle = \langle x, Pv\rangle = 0$.

(i) ⇒ (iii). By Pythagoras, $\|x\|^2 = \|Px\|^2 + \|x - Px\|^2 \geq \|Px\|^2$, so $\|P\| \leq 1$; with $\|P\| \geq 1$, $\|P\| = 1$.

(iii) ⇒ (i). Let $u \in R$, $v \in N$ and $t \in \mathbb{R}$. Then $P(u + tv) = u$, so $\|u\| \leq \|u + tv\|$, i.e.

$$0 \leq 2t\,\mathrm{Re}\,\langle u, v\rangle + t^2\|v\|^2 \quad \text{for all } t \in \mathbb{R}.$$

Dividing by t and letting $t \to 0^+$ and $t \to 0^-$ gives $\mathrm{Re}\,\langle u, v\rangle = 0$. Replacing v by $iv \in N$ gives $\mathrm{Im}\,\langle u, v\rangle = 0$. So $u \perp v$.